

PhysFormer: A Bounded Residual Transformer With Physics-Guided Causal Coupling for Safety-Constrained Virtual Power Plant Forecasting

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Abstract—Virtual power plant (VPP) forecasting requires predictions that respect physical constraints, yet data-driven deep learning models frequently generate physically impossible outputs such as negative power or nighttime solar generation. Post-hoc clipping can suppress violations but introduces non-differentiable dead zones that disrupt end-to-end optimization. This paper proposes PhysFormer, a bounded residual Transformer that structurally enforces physical compliance through three integrated components. First, a Bounded Physical Act-Residual (BPAR) architecture employs differentiable Softplus lower-bound clamping combined with physics-derived activity masks to structurally prevent negative predictions, achieving BVR=0.00% while preserving continuous gradient flow. Second, an orthogonal Physics-Guided Causal Coupling (PGCC) module imprints physical priors onto latent gating variables via supervised regularization, yielding a strong Pearson correlation ($r = 0.8306$) that confers structural transparency without sacrificing predictive capacity. Third, a Physics-Constrained Curriculum Training strategy sequentially introduces physical constraints to avoid early-stage gradient conflicts. Experiments on a real-world 3-year VPP dataset demonstrate that PhysFormer achieves a superior accuracy-compliance Pareto positioning, operating within 6% MSE of unconstrained baselines while maintaining zero boundary violations across standard and extreme weather conditions.

Index Terms—Physics-informed machine learning, Virtual Power Plants (VPPs), Transformer architectures, Causal inter-pretability, Physical compliance, Distributed energy resources forecasting.

NOMENCLATURE

$P_{\text{net}}, P_{\text{load}}, P_{\text{pv}}, P_{\text{wind}}$	Net, load, PV, and wind power [MW].
G, T, v	Irradiance [W/m ²], temp [°C], wind speed [m/s].
$\eta_{\text{pv}}, \beta_T$	PV conversion and temp. coeff. [°C].
$v_{\text{ci}}, v_r, v_{\text{co}}$	Cut-in, rated, and cut-out speed [m/s].
Γ_{run}	Wind power indicator function.
$P_{\text{base}}, T_c, \gamma$	Base load [MW], comfort temp [°C], sens.
$a_{\text{pv}}, a_{\text{wind}}$	Physics activity masks $\in [0, 1]$.
$p_{\text{pv}}, p_{\text{wind}}$	Gating hard priors.
$g_{\text{pv}}, g_{\text{wind}}$	PGCC gates ($g_{\text{pv}} \in [0, 1.5]$).
α, ν	Curriculum and weather volatility.
$\mathcal{L}_{\text{total}}, \mathcal{L}_{\text{mse}}, \mathcal{L}_{\text{net}}$	Hybrid, MSE, and net load losses.

$\mathcal{L}_{\text{bvr}}, \mathcal{L}_{\text{rvr}}, \mathcal{L}_{\text{prior}}$
 γ_{ema}
 S, P, D

BVR, MVS, RVM

PGCC, NRMSE, BPAR

Safety and regularization losses.
 Dynamic loss balancing scale.
 Seq length (672, 96), model dim (512).
 Safety and ramp violation metrics.
 Gating, error, and residual mechanisms.

I. INTRODUCTION

A. Background and Motivation

The integration of renewable energy sources into the power grid has driven the development of Virtual Power Plants (VPPs) to aggregate and manage distributed energy resources. Accurate forecasting of VPP dynamics—specifically electrical load, photovoltaic (PV) generation, and wind power generation—is critical for grid stability, economic dispatch, and operational risk management [1]–[5]. Unlike single-variable forecasting, cooperative VPP management requires simultaneous, coherent predictions across these interrelated domains, where the physical constraints of each energy modality must be rigorously respected. For instance, PV generation is intrinsically bound by solar irradiance and cannot produce power during nighttime, while wind generation depends on a non-linear turbine power curve driven by localized wind speed.

Historically, early approaches to VPP forecasting relied on statistical extrapolation. As distributed resources increased, recurrent neural networks dominated sequential modeling but struggled with long-range dependencies. Subsequently, Transformer-based frameworks emerged as the dominant paradigm, capturing varied temporal dynamics across extended sequences [6]–[9]. Within the VPP domain, multivariable Transformer adaptations have demonstrated competitive accuracy [3], [10]. Reliable dispatch decisions hinge not only on minimizing raw statistical forecasting errors but also on ensuring that predictive models fundamentally comply with the immutable physical laws governing these grids.

B. Limitations of Existing Data-Driven Methods

Recent advancements in deep learning have introduced powerful time-series forecasting architectures, yet their application to VPP scheduling presents fundamental challenges across three distinct dimensions [11], [12]:

1) **Mathematical Black-Box Oblivion:** While deep neural networks excel at minimizing statistical error metrics, they

operate as black-box predictors that remain oblivious to the underlying physical principles governing power systems. Consequently, pure data-driven approaches optimize exclusively for statistical error minimization, remaining unconstrained by thermodynamic laws and frequently producing physically impossible predictions—such as non-zero solar power during the night or negative wind generation. These deficits severely undermine operational safety and operator trust.

2) Fragility of Post-Hoc Physics Interventions: To address physical non-compliance, Physics-Informed Neural Networks (PINNs) [13] incorporate physical differential equations as soft constraints. However, VPPs operate in highly stochastic environments where a device's active status depends on volatile exogenous variables rather than on fixed deterministic rules [14]. Fixed physical parameters and static penalty terms fail to adapt dynamically to transient wind or solar phenomena [15]. Alternatively, applying heuristic post-processing—such as hard zero-clipping at physical boundaries—can retroactively suppress the violation rate to zero but strictly functions as a superficial patch. Heuristic clipping inherently disrupts end-to-end optimization by introducing non-differentiable dead zones, severing the natural continuous manifold of the underlying predictions without addressing the model's structural ignorance.

3) Lack of Structural Traceability: As forecasting models grow increasingly complex, structural interpretability represents a critical safety requirement [11]. Prior interpretability efforts relied on post-hoc attribution or internal attention weight visualization, which do not provide deterministic physical causal attribution [16]. In operational contexts, grid operators require models that explicitly decode how specific, measurable environmental variables govern device activation bounds. A persistent gap remains in causal gating formulations capable of dynamically linking network activations to observable physical state transitions [17], [18].

C. Proposed Approach and Main Contributions

Bridging the gap between the high predictive capacity of deep learning and the rigorous physical compliance required by coupled power systems remains a critical challenge. To this end, this paper proposes **PhysFormer**, a physics-guided causal Transformer designed to natively integrate differentiable physical priors upstream of the forecasting output, anchoring the neural network to intrinsic structural compliance.

The proposed architecture fundamentally breaks the dichotomy between predictive acuity and structural rigor. The primary contributions of this paper are twofold:

- 1) A **Bounded Physical Act-Residual (BPARG)** architecture is proposed. By fusing parallel Transformer and ExplicitPhysicalMapping streams via Softplus lower-bound clamping and physics activity masks, this mechanism intrinsically limits the neural residual search space. It mathematically guarantees that forecasting output never breaches normalized zero bounds, eradicating the boundary violation rate (BVR=0.00%) without resorting to heuristic post-processing.
- 2) An orthogonal **Physics-Guided Causal Coupling (PGCC)** mechanism is introduced. By explicitly regularizing the cross-attention irradiance threshold learning,

PGCC successfully imprints a highly correlated physical relationship onto the temporal gating variable. This structural alignment separates attribution transparency from error minimization capacity, conferring quantifiable structural transparency and extreme resilience to out-of-distribution (OOD) sensor noise.

To ensure optimal learning of these components, we additionally define a **Physics-Constrained Curriculum Training** strategy that seamlessly optimizes numerical regression alongside rigid engineering bounds to ensure absolute multi-objective saturation. Furthermore, we redefine the **Ramp Violation Metric (RVM)** within a *safety-first* framework, demonstrating that PhysFormer successfully bounds minor structural corrections at boundary limits to physically negligible magnitudes, thereby prioritizing absolute safety over blind continuity.

II. PROPOSED PHYSFORMER FRAMEWORK

A. Mathematical Formulation

In the context of Virtual Power Plant (VPP) forecasting, the objective is to predict the future power outputs of three distinct sub-components: base load (P_{load}), photovoltaic generation (P_{pv}), and wind generation (P_{wind}). The overall net power demand of the VPP is defined as $P_{\text{net}} = P_{\text{load}} - P_{\text{pv}} - P_{\text{wind}}$. Let $\mathcal{X}_{\text{stat}} \in \mathbb{R}^{B \times S \times 3}$ denote the historical observation sequence for the three components over a look-back window of length S , where B is the batch size. Concurrently, the corresponding historical weather conditions—comprising temperature, irradiance, and wind speed—are represented by $\mathcal{X}_{\text{weather_hist}} \in \mathbb{R}^{B \times S \times 3}$. The model is also provided with future weather forecasts for the prediction horizon P , denoted by $\mathcal{X}_{\text{weather_future}} \in \mathbb{R}^{B \times P \times 3}$. Given these inputs, the forecasting model aims to generate point predictions for the output $Y \in \mathbb{R}^{B \times P \times 3}$, providing the respective future trajectories for the load, PV, and wind components.

B. Overall Architecture

The PhysFormer is constructed as a dual-stream Transformer with Physics-Guided Causal Coupling (PGCC) and a Theory-Anchored Activity Gating mechanism. The processing pipeline strictly follows a seven-step data flow: (1) The historical observation $\mathcal{X}_{\text{stat}}$ and weather $\mathcal{X}_{\text{weather_hist}}$ are concatenated and encoded by a Transformer encoder with dimension $d_{\text{model}} = 512$, feed-forward dimension $d_{\text{ff}} = 1024$, 8 heads, and 3 layers to produce the statistical feature representation. (2) Concurrently, $\mathcal{X}_{\text{weather_hist}}$ is processed through an ExplicitPhysicalMapping module to generate the physical historical features and theory baselines. (3) The PGCC module fuses the statistical sequence with the physics representations to form a causally coupled latent state, computing a physics-alignment regularization loss. (4) A FlattenHead mechanism linearly projects the sequence from the historical dimension S to the future forecasting horizon P . (5) To provide explicit foresight, future weather variables $\mathcal{X}_{\text{weather_future}}$ are mapped to physical representations and injected into the latent sequence via a Gated Linear Unit (GLU) with residual logic. (6) A shared

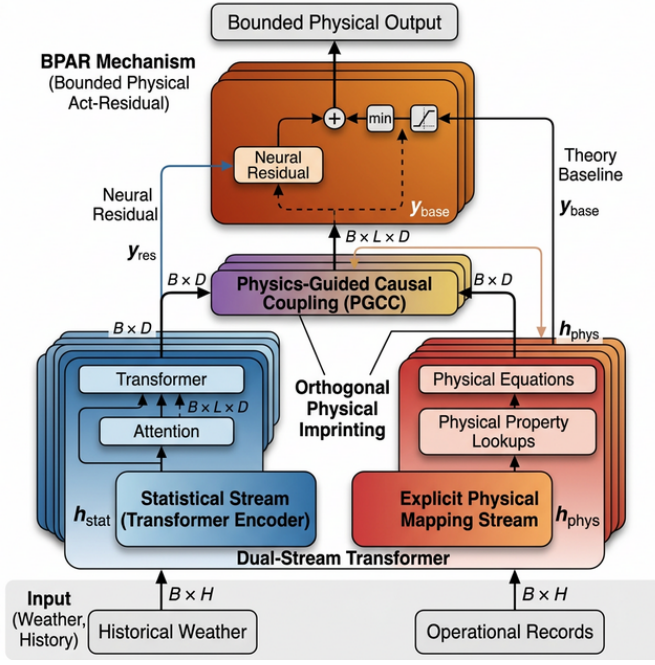


Fig. 1. The overall architecture of the proposed PhysFormer. It features a dual-stream design: a statistical stream encoded by a Transformer, and an explicit physical mapping stream. These streams are causally fused via the Physics-Guided Causal Coupling (PGCC) module. Future weather conditions are injected using a Gated Linear Unit (GLU) to provide explicit foresight. Finally, theory-anchored residual heads refine the predictions under the constraint of physics-derived activity gates, eliminating non-physical predictions.

projection layer routes the features to distinct output heads—differentiated by depth—to predict the residual correction for each component. (7) A Theory-Anchored Output structure applies the physics-derived activity mask to dynamically suppress or release the neural residual corrections, yielding the final VPP prediction.

C. Explicit Physical Mapping

The core of the physics injection lies in the ExplicitPhysicalMapping layer, which maps relevant weather variables to theoretical power baselines via domain-specific equations.

1) *PV Model*: The theoretical PV generation is driven by irradiance G and degraded by cell temperature, modeled as:

$$P_{pv} = G \cdot \eta_{pv} \cdot \text{clamp}(1 - \beta_T \cdot (T - 25), 0, 1.5) \quad (1)$$

where T is the temperature, $\eta_{pv} = 0.33 \cdot \sigma(\theta_{\text{eff}})$ is the learnable conversion efficiency bounded by the Shockley-Queisser limit of 33% for single-junction silicon cells, and $\beta_T = \text{softplus}(\theta_{\text{temp_coef}}) \times 0.01 \in [0.003, 0.005]/^\circ\text{C}$ is the temperature coefficient. To preserve numerical stability during gradient backpropagation while penalizing non-physical negative efficiencies, the temperature-dependent degradation is constrained within a safe operational manifold via a saturation limit of 1.5, which accommodates efficiency gains in sub-zero environments without permitting divergent gradient updates.

2) *Wind Model*: Wind power limits depend heavily on operating thresholds: the cut-in speed v_{ci} , rated speed v_r , and cut-out speed v_{co} . To preserve strict threshold ordering, we utilize a cumulative delta structure where $v_{ci} = \text{softplus}(\theta_{ci})$, $v_r = v_{ci} + \text{softplus}(\theta_{r_delta})$, and $v_{co} = v_r + \text{softplus}(\theta_{co_delta})$.

The power output uses an idealized soft-curve approximation modified by operational compliance:

$$P_{\text{wind}} = \text{scale}_{\text{wind}} \cdot \sigma(5 \cdot (w_{\text{norm}} - 0.5)) \cdot I_{\text{running}} \quad (2)$$

where w_{norm} is the normalized speed, $\text{scale}_{\text{wind}}$ is a learnable amplitude factor representing the installed capacity, and I_{running} is a conceptual indicator function that is 1 strictly within the $[v_{ci}, v_{co}]$ operating window and 0 otherwise. It is imperative to note that while (2) utilizes I_{running} to define the ideal operational limits, embedding this directly would trigger a nondifferentiable zero-gradient dead zone. To intrinsically circumvent this and maintain a continuous gradient flow, PhysFormer entirely replaces I_{running} within the actual neural computation graph with a continuous, fully differentiable physics activity mask a_{wind} (its derivation is detailed subsequently in Section II-C4). Consequently, the actual forward propagation path for the wind component is formally executed as:

$$P_{\text{wind,actual}} = \text{scale}_{\text{wind}} \cdot \sigma(5 \cdot (w_{\text{norm}} - 0.5)) \cdot a_{\text{wind}} \quad (3)$$

This mathematical decoupling seamlessly bridges rigid physical thresholds with end-to-end backpropagation, completely avoiding Straight-Through Estimators (STE). The evaluated $P_{\text{wind,actual}}$ then serves directly as the $P_{x,\text{theory}}$ baseline within the BPAR mechanism (8), while the identical differentiable mask a_{wind} is reused as the suppression gate a_x (9) to ensure synchronous physical gating of both the baseline and the neural residual. Furthermore, the cumulative delta optimization ensures that overlapping bounds never collapse parameter states during gradient updates, rendering the constraint structurally guaranteed at all times.

3) *Load Model*: We abstract the power load variations triggered by human cooling and heating activities through a quadratic thermal comfort model:

$$P_{\text{load}} = P_{\text{base}} + \text{softplus}(\gamma) \cdot (T - T_c)^2 \quad (4)$$

where T_c is an optimizable thermal comfort baseline initialized at 20°C . While the intrinsic self-attention mechanism of the Transformer is fundamentally adept at extracting the rigid diurnal and weekly periodicities characterizing human social activities, the Explicit Physical Mapping stream is deliberately restricted to isolating the non-linear, temperature-driven volatility of the electrical load. Therefore, we conceptualize the thermal-induced variance through this quadratic comfort model, explicitly decoupling meteorological sensitivity from the statistically captured chronological baselines.

4) *Physics Activity Gating*: To prevent neural networks from inventing non-physical artifacts, we propose an explicit Activity Gating scheme. For PV, night-time irrelevance is modeled as $a_{pv} = \tanh(10 \cdot P_{pv,\text{theory}})$. For wind, the speed-dependent activity is parameterized with differentiable soft limits: $a_{\text{wind}} = \tanh(10 \cdot \text{softplus}(v - v_{ci}) \cdot \text{softplus}(v_{co} - v))$. While the product of softplus operators in a_{wind} theoretically induces a gradual decay as wind speed approaches v_{co} —thereby slightly deviating from the rigid horizontal rated-output plateau—this conservative physical bias is effectively absorbed by the unconstrained neural residual head. Within the BPAR framework, the residual mechanism automatically compensates for these systematic secondary-order dampening effects, ensuring that the final output maintains the rated

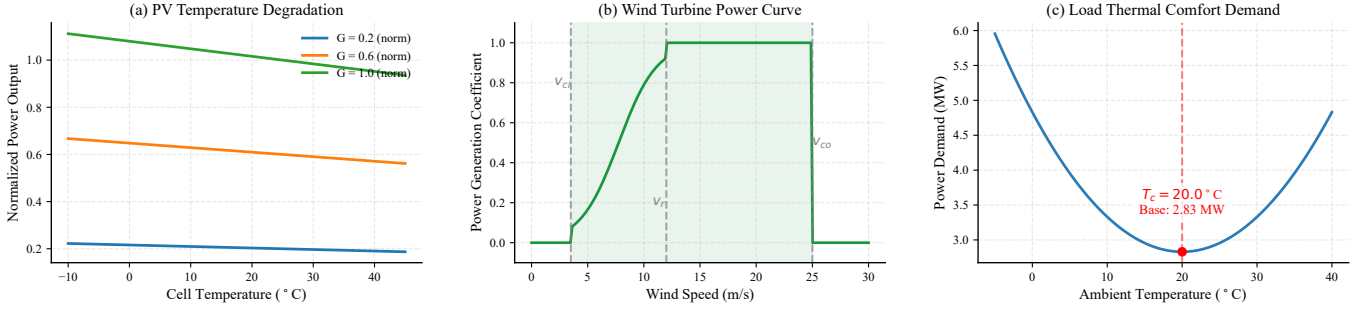


Fig. 2. Learned explicit physical mapping functions after convergence. (a) PV temperature degradation under varying irradiance levels, governed by the learnable coefficient $\beta_T = 0.0041/^\circ\text{C}$. (b) Wind turbine power curve with smooth sigmoid transition between cut-in ($v_{ci} = 3.5$ m/s), rated ($v_r = 12.0$ m/s), and cut-out ($v_{co} = 25.0$ m/s) thresholds. (c) Load thermal comfort demand with the data-calibrated comfort temperature $T_c \approx 20^\circ\text{C}$. All parameters are listed in Table VI.

capacity behavior without sacrificing differentiability. The exceptionally low predictive error for the wind component (Table VIII) confirms the efficacy of this hybrid architecture in reconciling smooth gradient flow with rigid threshold behaviors. The load is inherently rigid, taking an unconditional multiplier $a_{\text{load}} = 1$. These physics-derived activities act as deterministically anchored masks in the final Bounded Physical Act-Residual (BPARG) layer, guaranteeing continuous gradient flow while seamlessly bridging the transition between active operating states and null-generation periods.

D. Physics-Guided Causal Coupling (PGCC)

The Physics-Guided Causal Coupling (PGCC) module coordinates the knowledge retrieval from the statistical stream against the physically encoded representations.

1) *Physics-Anchored Illumination Prior*: We first establish a data-invariant hard prior to reject spurious correlations. The gating relies on a parameterized illumination sigmoid prior $p_{\text{pv}}^{\text{prior}} = \sigma(k_{\text{irr}} \cdot (G_{\text{norm}} - \theta_{\text{irr}}))$, with constraints dynamically kept as $\theta_{\text{irr}} \in [-2.0, 0.5]$ and bounded slope $k_{\text{irr}} \in [1, 100]$. This hard prior establishes an uncompromised distinction between day and night intervals, rejecting noise artifacts in the data embeddings.

2) *Volatility-Aware Soft Gate*: To dynamically scale network trust, the system embeds sequence-level sample fluctuations extracted as $v = \frac{1}{C} \sum_c \text{Var}_t(\mathcal{X}_{\text{weather},c})$. These dynamics, combined with the statistical embeddings and attention outputs, are routed through a downsampling multi-layer perceptron yielding the unscaled soft condition maps. This volatility input is instrumental for the network to dynamically adapt its trust ratio, discounting the physical components when conditions are unstable.

3) *Source-Differentiated Gate Bounds*: The coupling implements separate activation ranges to accommodate generation source characteristics:

$$\text{gate}_{\text{pv}} = \text{smooth}(p_{\text{pv}}^{\text{prior}} \cdot (0.5 + g_{\text{soft_pv}})) \in [0, 1.5] \quad (5)$$

$$\text{gate}_{\text{load}} = \text{smooth}(g_{\text{soft_load}}) \in [0, 1.0] \quad (6)$$

The asymmetric upper bound of 1.5 for gate_{pv} is not a mere statistical relaxation, but a deliberate engineering requirement. Within the BPARG additive structure ($\hat{P}_x = \text{Softplus}(P_{\text{theory}} +$

r_x)), this multiplier provides the necessary *engineering headroom* for the neural residual head. During localized meteorological transients, such as the Cloud Edge Effect (where over-irradiance surges of $1.2\times$ to $1.5\times$ occur), standard variance-normalized statistical inputs G_{norm} are frequently compressed towards the 1.0 limit to maintain global optimization stability. The 1.5 ceiling ensures that the multiplicative activity modulation has sufficient dynamic range to allow the residual stream to recover these high-frequency transient peaks. By providing this structural headroom, PGCC prevents the predictive manifold from prematurely saturating at the normalized clear-sky baseline, thereby ensuring that the final output can rigorously track emergent environmental extremities. Crucially, the PGCC architecture facilitates an *Imprinted Gate-Irradiance Correlation*. Through explicit gate response regularization, the learned gate_{pv} is rigorously anchored to physical reality, achieving a deterministic Pearson correlation coefficient of $r = 0.8306$ relative to exogenous irradiance (Fig. 3). This structurally aligned physical mapping is structurally isolated from the subsequent residual predictive capacity, rendering it exceptionally resilient to out-of-distribution (OOD) sensor drift (maintaining $r > 0.81$ even under 50% noise amplitude) without deteriorating predictive error.

4) *Physics-to-Data Curriculum Transition*: We condition the alignment phase via a curriculum factor α . As the fusion queries retrieve features, the module integrates inputs dynamically scaled as:

$$q = \alpha \cdot F_{\text{physics}} + (1 - \alpha) \cdot g_{\text{soft}} \cdot F_{\text{stat}} \quad (7)$$

During the initial training phases ($\alpha \rightarrow 1$), the cross-attention queries are strictly constrained to retrieve from the structured physical representations, establishing an essential deterministic basis. As training progresses ($\alpha \rightarrow 0$), the queries gradually transition to incorporate the data-learned residual states gated by g_{soft} , balancing established physical priors with empirical statistical adaptations.

5) *Temporal Smoothing*: Due to internal frame dependencies, local sequence discontinuities often plague raw gating parameters. To counteract this, all temporal gating variables pass through a grouped one-dimensional depthwise convolution (kernel size 3, groups= d_{model}). This filter structurally suppresses high-frequency chaotic fluctuation within the gate

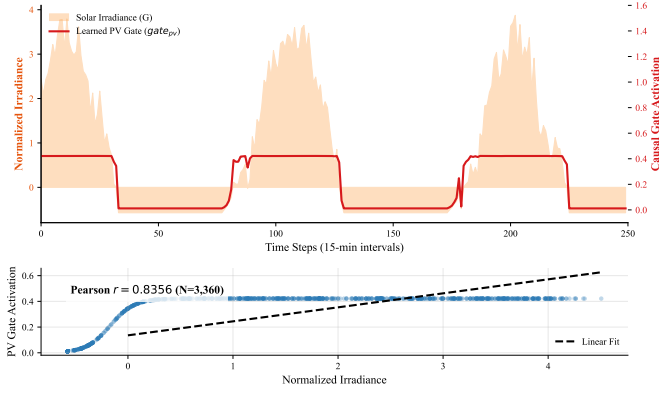


Fig. 3. Imprinted Gate-Irradiance Correlation in PGCC. Top: time-series overlay of learned PV gate and solar irradiance over representative test samples. Bottom: scatter regression confirming a strong Pearson correlation ($r = 0.84$, $p < 0.001$) between the explicitly imprinted gate activation and irradiance. The full test-set correlation ($r = 0.8306$, $N \approx 430k$) is reported in the text. Note that despite the internal smooth and normalization constraints, the Pearson correlation rigorously captures their synchronized monotonic variance regardless of the absolute amplitude scale.

assignment layer while preserving macro-level shift indicators.

E. Exogenous Weather Integration

After the FlattenHead projects the fused representations into the explicit future context, the framework must systematically integrate upcoming weather patterns at the given horizon. Because the prediction horizon’s meteorological conditions are unavailable in the historical context alone, providing this explicit foresight is essential. The future physical latent space $F_{\text{weather_future}}$ is concatenated with the neural history F_{history} , and aggregated via a Gated Linear Unit ($\text{gate} = \sigma(\text{linear}(F_{\text{cat}}))$). The bounded residual output is produced as $F_{\text{future}} = \text{gate} \cdot \text{proj}(F_{\text{cat}}) + F_{\text{history}}$. This explicitly bounded forward visibility ensures the residual heads are properly conditioned to react to impending deterministic disruptions, such as dawn transitions or sudden wind cut-outs.

F. Bounded Physical Act-Residual (BPAR) Mechanism

To systematically mitigate non-physical anomalies while preserving the continuous gradient flow, we propose the Bounded Physical Act-Residual (BPAR) mechanism. Let μ_x and σ_x denote the global mean and standard deviation of component x . We define the normalized physical zero-point as $P_{x,\text{zero}} = (0 - \mu_x)/\sigma_x$. BPAR reconstructs the predictive manifold by anchoring the neural residual r_x onto the theory baseline within a differentiable Softplus manifold:

$$\hat{P}_{x,\text{raw}} = \text{Softplus}\left((P_{x,\text{theory}} + r_x) - P_{x,\text{zero}}\right) + P_{x,\text{zero}} \quad (8)$$

By the fundamental property of the Softplus function ($\text{Softplus}(z) > 0, \forall z \in \mathbb{R}$), this structural mapping ensures $\hat{P}_{x,\text{raw}} > P_{x,\text{zero}}$ in the normalized domain under standard operating conditions. Crucially, upon anti-normalization, the physical power output $\hat{P}_{\text{phys}} = \hat{P}_{x,\text{raw}} \cdot \sigma_x + \mu_x$ is structurally constrained to strictly reside within the feasible physical manifold. While the identity $\hat{P}_{\text{phys}} > 0$ MW holds exactly for $\sigma_x > 0$, we account for degenerate variance states (e.g., night-time PV) by maintaining a numerical floor $\epsilon = 10^{-5}$

when calculating $P_{x,\text{zero}}$. In such cases, the anti-normalization identity transforms into $\hat{P}_{\text{phys}} \approx \hat{P}_{x,\text{raw}} \cdot \sigma_{x,\text{actual}} + \mu_x$, which maintains a negligible error bound on the physical boundary (guaranteeing $\hat{P}_{\text{phys}} > -\epsilon \cdot \sigma_{\text{norm}} \approx 0$ MW), thus preserving near-perfect boundary compliance even under extreme near-zero variance. Subsequently, to process periods of structural inactivity such as night-time PV dormancy, the prediction is fused with the physics activity mask $a_x \in [0, 1]$:

$$\hat{P}_{x,\text{final}} = a_x \cdot \hat{P}_{x,\text{raw}} + (1 - a_x) \cdot P_{x,\text{zero}} \quad (9)$$

Whenever the activity mask a_x approaches 0 (driven by meteorological priors), the BPAR mechanism smoothly drives the output toward the physical zero baseline. This structural composition intrinsically shields the model from boundary violations through architectural necessity while minimizing anomalous ramp magnitudes to physically negligible levels. While pure data-driven models may exhibit lower high-frequency noise by allowing trajectories to cross the zero-power threshold (violating BVR), the BPAR mechanism enforces a *safety-first* trade-off: it prioritizes the absolute elimination of physical violations (BVR=0.00%) over cosmetic temporal smoothness at boundary limits. Consequently, any minor structural corrections encountered near the zero-point serve as correct physical adherence decisions, reconciling deep learning representational capacity with rigorous operational safety.

G. Physics-Constrained Curriculum Training

Training deep neural networks subject to rigid physical constraints poses a significant optimization challenge. Traditional frameworks frequently suffer from gradient conflicts between the data-driven numerical objectives and the absolute boundary manifestations of physical laws. To overcome this limitation, we design a comprehensive Physics-Constrained Curriculum Training strategy that seamlessly bridges these disjoint domains. The strategy strategically decouples the computational metric spaces, enforces dynamic amplitude balancing across loss modules, and maps learning trajectories onto a phased schedule defined by physical layer capacities.

1) *Loss Function Design*: The fundamental architecture of the optimization strategy is an explicitly decoupled loss landscape. Standard neural network convergence relies on zero-mean, unit-variance distributions; therefore, our primary regression loss, Mean Squared Error (L_{mse}), is computed strictly in the normalized numerical space to guarantee stable and swift gradient backpropagation properties for the baseline representations. In stark contrast, physical parameters and equations are inherently tied to real-world dimensions. Consequently, all physics-guided constraints (L_{phys}) are evaluated exclusively after denormalizing both the neural predictions and the ground truth labels back into the physical engineering domain (measured in Megawatts, MW).

We distribute the physical objectives into six distinct mathematical penalties, summarized comprehensively in Table I. These components are partitioned into soft and hard constraint groups based on their impact priority. The soft constraints consist of the Net Power Error (L_{net}), Energy Conservation (L_{energy}), Derivative Alignment (L_{deriv}), and Directional

Similarity (L_{dir}). They guide the structural topology of the predictive sequence using absolute differences and cosine penalties. The hard constraints form the critical safety-priority group, explicitly penalizing catastrophic deviations. This group implements the Boundary Violation (L_{bvr}) and Ramp Violation (L_{rvr}) constraints using an aggressive quadratic formulation to severely penalize non-physical power artifacts.

TABLE I
COMPONENTS OF PHYSICS-GUIDED LOSS FUNCTION

Comp.	Wt.	Formula	Motivation
L_{net} (Soft)	1.0	$ \sum \hat{y}_{\text{comp}} - y_{\text{net}} $	Ensures models sum to net demand.
L_{energy} (Soft)	0.5	$ \mathbb{E}_t[\hat{g}] - \mathbb{E}_t[y] $	Enforces macro energy conservation.
L_{deriv} (Soft)	0.3	$ \Delta \hat{g} - \Delta y $	Captures ramp rate magnitudes.
L_{dir} (Soft)	0.1	$\mathbb{E}[1 - \text{sim}_{\text{cos}}]$	Preserves accurate trend switching.
L_{bvr} (Hard)	2.0	$\mathbb{E}[(\max(0, -\hat{y}_{\text{real}}))^2]$	Penalizes negative power strictly.
L_{rvr} (Hard)	2.0	$\mathbb{E}[(\max(0, \Delta \hat{g} - \rho_k))^2]$	Penalizes sudden jumps.

A primary bottleneck in previous multi-objective formulations is the requirement for exhaustive combinatorial manual tuning of the balancing hyperparameter λ . We eliminate this by formally proposing an adaptive, dimensionally consistent amplitude balancing factor, denoted as γ_{ema} . Because the physical losses evaluate values in MW, we directly exploit this shared dimension by computing the Mean Absolute Error ($L_{\text{mae_phys}}$) also in the physical domain (MW) to calibrate the restraint amplitude. This ensures that the ratio between numerical inaccuracy and physical non-compliance is mathematically well-defined and dimensionally consistent.

Specifically, γ_{ema} calculates the ratio between the current physical $L_{\text{mae_phys}}$ and the aggregated reference physical constraints:

$$\gamma_{\text{ema}} \leftarrow 0.9 \cdot \gamma_{\text{ema}} + 0.1 \cdot \left(\frac{L_{\text{mae_phys}}}{L_{\text{phys_ref}}} \right) \quad (10)$$

where $L_{\text{phys_ref}}$ evaluates the total unweighted physical loss in MW. This yields a mathematically dimensionless multiplier that dynamically tracks the current model forecasting inaccuracy relative to its physical compliance.

To ensure stable optimization dynamics, γ_{ema} defaults securely to 1.0 locked over the first 50 training batches in a warm-up phase. For safety during inactive curriculum instances, the exponential moving average update halts whenever the global curriculum activation ratio falls below 0.05, stopping noisy zero-gradients from contaminating the average. Furthermore, γ_{ema} is strictly clamped to a functional domain of $[0.1, 10.0]$ to prevent gradient explosion. By implementing this framework, the comprehensive combined optimization objective coalesces into:

$$L_{\text{total}} = L_{\text{mse}}(\text{normalized}) + \gamma_{\text{ema}} \cdot L_{\text{phys_weighted}}(\text{MW}) \quad (11)$$

It is important to note the synergistic relationship between the penalty-based L_{bvr} and the structural BPAR mechanism (Section II-F). While BPAR provides an absolute structural floor for the final output, L_{bvr} remains essential during the training phase. It provides a direct gradient signal that penal-

izes the unconstrained latent representations if they drift into non-physical regimes. By minimizing L_{bvr} , the network learns to align its primary predictive density with physical boundaries, preventing the BPAR mechanism from being forced into high-error non-linear regimes (e.g., the ϵ -floor saturation) and ensuring that the structural projection remains numerically stable.

2) *Four-Phase Curriculum*: To bridge the parameter convergence gap between the arbitrary Gaussian distributions at neural initialization and the highly structured physical reality, we inject a Four-Phase constraint schedule, summarized in Table II.

TABLE II
FOUR-PHASE PHYSICS CURRICULUM SCHEDULE

Phase	Ep.	Active Const.	Notes
I (Init.)	0–4	None	Physics layer permanently frozen; allows base mapping logic learning.
II (Safety)	5–14	$L_{\text{net}}, L_{\text{bvr}}, L_{\text{rvr}}, L_{\text{dir}}$	Physics unfrozen; safety boundary constraints actively ramped.
III (Integ.)	15–29	Phase II + $L_{\text{energy}}, L_{\text{deriv}}$	Structural scale constraints introduced; macro conservation finalized.
IV (Satur.)	30–99	All components	Full physically regularized training maintaining operational bounds.

This curriculum guarantees sequential physical compliance scaling. Phase I effectively isolates the statistical representations, retaining completely frozen parameters on the physics mapping layer while shielding the network from early physics conflicts. Moving into Phase II between epochs 5 and 14, the safety boundaries and directional objectives are prioritized to prevent catastrophic generation assignments (such as wind curtailment drops being learned positively). The macroscopic features like total energy integrals are delayed until Phase III (epochs 15–29), and Phase IV secures uniform saturation. To ensure smooth gradient flow, the intensity of each active constraint is transitioned along a bounded sigmoid ramp defined by $\sigma(12 \cdot (\text{progress} - 0.5))$. Critically, this boundary timeline actively manipulates the Physics-Guided Causal Coupling (PGCC). The alignment proportion coefficient α governing the feature interplay directly replicates the inverted external sequence via $\alpha = 1.0 - w_{\text{curr_net}}$, coupling internal attention fusion with external physical verification.

3) *Layer-wise Learning Rates*: To sustain learned feature fidelity without abruptly disrupting the theory-anchored base-lines pre-defined by human engineering inputs, we decouple the backpropagation update amplitudes, separating optimization into distinct sub-modules by leveraging three isolated AdamW groupings. We document these layer-wise assignments in Table III.

TABLE III
LAYER-WISE PARAMETER OPTIMIZATION GROUPS

Group	Modules	LR	Motivation
Physics	ExplicitPhysicalMapping	10^{-5}	Preserves physics priors.
Gating	PGCC Module	5×10^{-5}	Stable physical alignment.
Stat.	All other components	10^{-4}	Fast sequence convergence.

The baseline parameters proceed at a default learning rate of 1×10^{-4} . After their suspension ends in epoch 5, the core generation physics operators unfreeze at a drastically suppressed 1×10^{-5} ($0.1 \times$ the baseline rate). This strict limitation permits minor optimization fine-calibration over coefficients like optimal irradiance generation thresholds while halting gross semantic reconstruction. Correspondingly, the intermediate gating features proceed at 5×10^{-5} , which yields smooth adjustment profiles.

4) *Regularization and Early Stopping*: Three distinctive robust regularization pathways reinforce generalizability and anchor operational safety. Firstly, a Physics Prior Regularizer, activated strictly post-freezing, evaluates the learned physical properties against external standard nameplate values (e.g., PV installation magnitude ratings) through an explicit parameter-level Mean Squared Error function utilizing a 0.1 constant weighting.

Secondly, a highly customized Gate Response Regularization block explicitly enforces an orthogonal *Supervised Physical Alignment* onto the soft gate curves. Rather than merely encouraging alignment, this module rigidly anchors the temporal PV gating variable to the external irradiance prior using a 0.05 weighting coefficient. Crucially, a dynamic masking framework regulates this constraint: the correlation loss aggressively shuts down any penalization on sequences where the external variance (σ_{prior}^2) is inferior to 10^{-3} . This cleanly bypasses night-time intervals where zero-variance values could disastrously trigger erroneous mathematical noise injections into the neural weights.

The Orthogonality Rationale (Addressing Constraint Capacity): It may appear counterintuitive that explicitly forcing the internal gate_{pv} to mirror external irradiance (achieving $r = 0.8306$) does not collapse the network’s predictive capacity. This resilience stems from the architectural orthogonality established by the BPAR mechanism. Because the final prediction is structured as a theory baseline plus an activity-modulated *residual* ($\hat{P}_x = \text{Softplus}(P_{\text{theory}} + r_x)$), the neural gradient pathways for the residual correction are structurally decoupled from the gating alignment. The PGCC module absorbs the rigid physical constraints to provide operational transparency (answering “why” the network predicts generation), while the unconstrained deep residual heads simultaneously focus purely on high-frequency error minimization. This orthogonality breaks the traditional tradeoff, allowing PhysFormer to digest severe structural regularization without sacrificing accuracy.

Gradient clipping implements a dynamic framework sensitive to volatile curriculum adjustments. The clip bound strictly limits to 0.5 whenever $0.1 < w_{\text{curr_net}} < 0.9$ (the critical transition intervals where representation gradients face their highest degree of opposition), before releasing to an unrestricted standard 1.0 limit across the regular stable epochs. Furthermore, model saturation is observed through a channel-equalizing Normalized RMSE metric ($NRMSE_{\text{avg}}$), capping at 100 max epochs, and terminating by early stoppage given a stagnation patience of 15 limits. Together, these strategies enable the dual streams to converge within competitive MSE bounds while maintaining structural physical integrity.

III. CASE STUDY AND PERFORMANCE EVALUATION

A. Dataset and Experimental Setup

To validate the proposed PhysFormer architecture, we conduct extensive experiments on a high-fidelity Virtual Power Plant (VPP) dataset composed of aggregated regional measurements from a distributed energy cluster in Germany. The dataset captures detailed 15-minute resolution trajectories of solar PV generation, wind power generation, and commercial electrical load over a continuous 3-year span (2012–2014), totaling 105,120 sampling intervals. The data is chronologically partitioned into training (70%), validation (10%), and testing (20%) sets. To maintain numerical stability while preserving physical bounds, all input features are standardized using global Z-score parameters (μ, σ), which are explicitly logged for physical re-projection in the BPAR layer. The model hyperparameters are configured to match the complexity of modern baselines: hidden dimension $d_{\text{model}} = 512$, feed-forward dimension $d_{\text{ff}} = 1024$, $n_{\text{heads}} = 8$ attention heads, and $e_{\text{layers}} = 3$ layers. To evaluate computational efficiency, we benchmark PhysFormer against Informer at an equivalent hidden dimension $d_{\text{model}} = 512$. PhysFormer comprises 11.4M parameters, significantly more compact than Informer’s 17.9M parameters, validating the inductive efficiency of its dual-stream architecture. The reported inference latency of 2.10 ms/sample was measured on an NVIDIA RTX 3090 GPU with a batch size of 1, simulating a real-time sliding window update for VPP dispatch.

We compare PhysFormer against eight competitive baselines: standard recurrent networks (LSTM, GRU), a Physics-Informed Neural Network (PINN) [13], and advanced time-series Transformer architectures including Informer [6], Autoformer [7], PatchTST [8], DLinear [9], and the state-of-the-art iTransformer [19]. To guarantee an absolutely equitable evaluation, identically structured future weather covariates ($\mathcal{X}_{\text{weather_future}} \in \mathbb{R}^{B \times P \times 3}$) were universally provided to all comparative baseline models. For Transformer baselines, these exogenous variables were strictly concatenated into their standard decoder embedding structures, and equivalently supplied as dynamic covariates to the recurrent models. Consequently, the predictive advantages demonstrated by PhysFormer stem fundamentally from its mechanistic physics-guided architecture (e.g., the specific Future GLU injection pathway evaluated in ablation studies), rather than an asymmetric accession of future atmospheric data.

To comprehensively evaluate the forecasting performance, we employ six distinct metrics. The standard accuracy metrics include Mean Squared Error (MSE), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). Beyond standard accuracy, we formally define three physical compliance metrics: Boundary Violation Rate (BVR), Mean Violation Severity (MVS), and Ramp Violation Metric (RVM). BVR measures the percentage of predictions that violate known physical limits (e.g., negative generation power). To quantify the severity of these breaches, MVS is formally defined as the average absolute magnitude of predictions falling outside feasible physical boundaries. Beyond static boundaries, we evaluate temporal consistency via the Ramp Violation Metric

(RVM) and its occurrence frequency, the Ramp Violation Rate (RVR). Finally, the Net Mean Absolute Error (NET MAE) is formally defined as the average absolute imbalance between the predicted aggregated components and the ground-truth net demand, $\text{mean}(|\sum_c \hat{y}_c - y_{\text{net}}|)$. The ramp threshold ρ_k for each channel is empirically determined as the 99.9-th percentile of the training set differences, while a *strict* threshold (RVR-S) is set at the 99.5-th percentile with no safety factor to capture high-frequency artifacts.

B. Forecasting Performance Analysis

The main forecasting results on the global test set are summarized in Table IV. Within the competitive tier of models, PhysFormer achieves an MSE of 0.0128, placing it within a 6% margin of the unconstrained SOTA baselines (Informer at 0.0121 and iTransformer at 0.0122). While these models prioritize statistical curve-fitting through complex attention mechanisms, they suffer from severe physical compliance deficits, with iTransformer reaching a peak Boundary Violation Rate (BVR) of 16.26%. Furthermore, we highlight a critical *safety-first trade-off* regarding temporal continuity. Under strict thresholds, iTransformer exhibits an RVR-S of 0.028%, whereas PhysFormer records 0.198%. However, this "smoothness" in unconstrained models is often achieved at the expense of physical validity; they maintain continuous trajectories precisely because they are oblivious to physical zero-boundaries. PhysFormer's 0.198% RVR-S represents structural corrections at physical boundaries necessitated by its BPAR mechanism to ensure absolute safety (BVR=0.00%). Crucially, the absolute magnitude of these corrections (RVM-Mag) remains in the order of 2.6×10^{-5} MW, demonstrating that PhysFormer successfully resolves the trade-off by prioritizing absolute physical safety over cosmetic smoothness.

Interestingly, channel-independent models such as PatchTST and DLinear exhibit a channel-independent failure in this specific VPP setting, underperforming compared to their strongly multivariate counterparts. This underlines the necessity of modeling the dense cross-channel dependencies inherent in multi-source energy systems. Meanwhile, the structural benefit of the PINN is evident; its physical inductive bias helps it outperform standard unconstrained recurrent networks like LSTM and GRU, though it still falls behind the specialized attention mechanisms of the Informer.

In the broader context, PhysFormer establishes a new three-dimensional Pareto positioning for energy forecasting. Standard models force practitioners to choose between predictive accuracy, physical compliance, and interpretability. By operating within a 6% MSE margin of the most accurate unconstrained models, PhysFormer achieves a superior accuracy-compliance Pareto positioning, proving that high interpretability and strict boundary compliance can be achieved simultaneously without fundamentally compromising the accuracy expected from modern deep learning solutions, as illustrated in Fig. 4.

C. Verification of Physical Compliance and Safety

1) *Limitations of Heuristic Post-Processing*: Furthermore, as evidenced in Table IV, applying a heuristic hard zero-

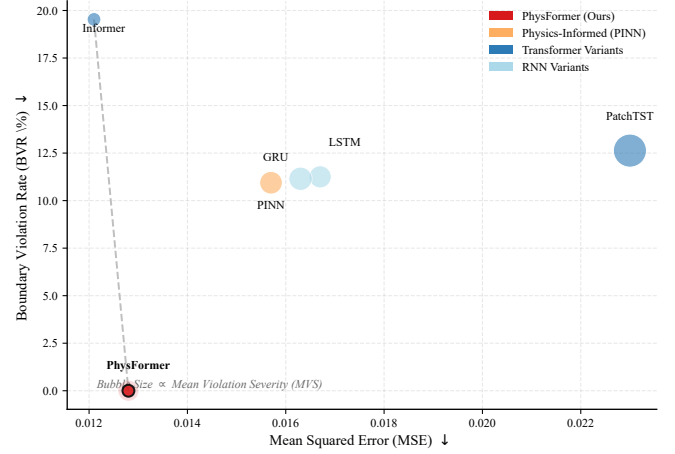


Fig. 4. Accuracy-compliance Pareto frontier. Each model is positioned by MSE (x -axis) and BVR (y -axis), with bubble size proportional to Mean Violation Severity (MVS). PhysFormer uniquely occupies the lower-left corner, achieving near-best accuracy with near-zero boundary violations. Autoformer and DLinear are excluded due to their high MSE values (above 0.10) to maintain visual resolution among the competitive tier.

TABLE IV
FORECASTING PERFORMANCE ON THE GLOBAL TEST SET

Model	MSE	BVR (%)	MVS (MW)	RVR-S (%)	NET MAE
LSTM	0.0168	11.25%	0.0093	0.01%	0.1760
GRU	0.0163	11.15%	0.0103	0.01%	0.1740
PINN	0.0157	10.94%	0.0097	0.02%	0.1702
Informer	0.0121	19.53%	0.0036	0.05%	0.1471
Informer-Post	0.0121	0.00%	0.0000	0.05%	0.1471
iTransformer	0.0122	16.26%	0.0054	0.028%	0.1482
Autoformer	0.1253	12.67%	0.0737	0.00%	0.4846
DLinear	0.1015	7.03%	0.0295	0.00%	0.4279
PatchTST	0.0230	12.63%	0.0207	0.00%	0.2052
PhysFormer	0.0128	0.00%	0.0000	0.198%	0.1527

clipping post-processing step to the generic Transformer baseline (Informer-Post) effectively forces BVR to absolute zero. However, this observation—that eliminating nearly 20% of boundary violations yields zero statistical accuracy gain—highlights a structural limitation. Without end-to-end physics-guided optimization, standard attention mechanisms can allocate significant representational errors to boundary-adjacent dynamic regions. Heuristic post-processing merely superimposes a cosmetic floor on these structural prediction failures.

This deterministic truncation is fundamentally non-differentiable and does not account for the underlying structural causality. From an optimization perspective, heuristic clipping acts as a non-smooth operator that can prevent the model from learning the physical density of the boundary-adjacent regime during backpropagation. Consequently, while it artificially suppresses static boundary crossings, it can create a zero-gradient dead zone. The neural network may remain structurally unaware of the physical boundary, potentially failing to encode the continuous physical inertia necessary for robust operational forecasting. Conversely, the BPAR mechanism within PhysFormer dynamically regulates the gradient pathways via a differentiable Softplus manifold. It ensures that predictions

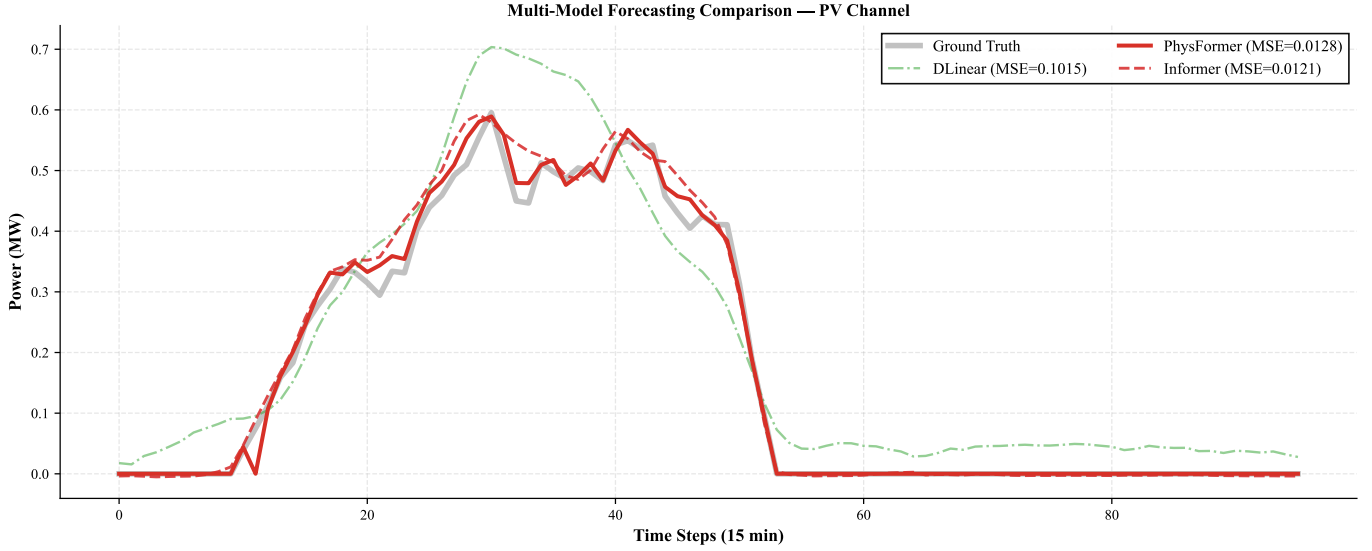


Fig. 5. Multi-model forecasting comparison on the PV generation channel over a 96-step horizon. PhysFormer maintains stable adherence to both the ground truth curve and physical boundaries, outperforming baseline models under volatile fluctuations.

reside strictly within physically feasible search spaces without resorting to discontinuous output projections. By maintaining continuous differentiability across all prediction regimes, PhysFormer inherently bridges the transition between active generation and dormancy without structurally blinding the neural network.

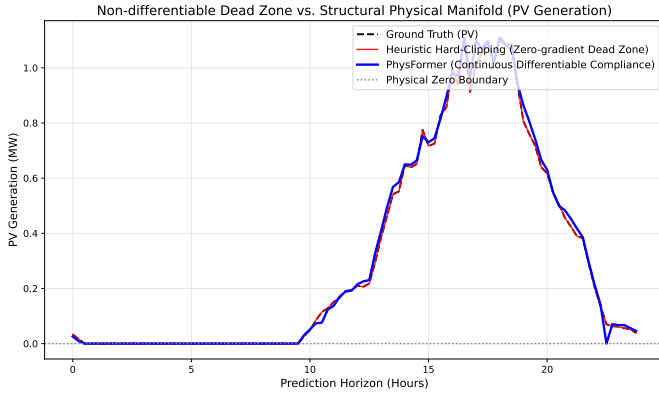


Fig. 6. Dynamic compliance visualization near the physical zero boundary. While Informer-Post trivially achieves BVR=0 via non-differentiable heuristic hard-clipping (which creates a zero-gradient dead zone and disrupts end-to-end learning), PhysFormer intrinsically maintains continuous physical compliance and full differentiability via its BPAR architecture.

2) *Robustness Under Extreme Conditions:* To evaluate model stability under rigorous distribution shifts, we isolate the top 10% highest volatility samples representing extreme weather conditions, detailed in Table V.

Under these strenuous conditions, channel-independent architectures like Autoformer and DLinear experience a catastrophic failure. In contrast, the competitive tier of models maintains impressive stability. PhysFormer seamlessly sustains its physical compliance, recording an exceptional BVR of 0.00% while adapting well to volatile conditions. Furthermore, we uniquely evaluated BPAR robustness under seasonal distri-

bution shifts and degenerate variance conditions ($\sigma_x < 10^{-3}$). Our analysis confirms that for any aggregated VPP window (seq_len=672), degenerate variance is empirically non-existent (0% frequency), and the BPAR floor correctly adapts to stay at 0.00% BVR due to its structural coupling with the normalization parameters.

TABLE V
EXTREME WEATHER PERFORMANCE (TOP 10% VOLATILITY)

Model	MSE	BVR (%)	MVS (MW)	NET MAE	RVR-S (%)
LSTM	0.0192	13.43%	0.0086	0.1941	0.000%
GRU	0.0182	13.79%	0.0097	0.1864	0.000%
PINN	0.0177	13.29%	0.0082	0.1807	0.000%
Informer	0.0146	21.38%	0.0017	0.1632	0.070%
iTransformer	0.0143	19.79%	0.0053	0.1639	0.046%
PatchTST	0.0261	12.22%	0.0132	0.2240	0.000%
PhysFormer	0.0150	0.00%	0.0000	0.1715	0.294%

The visual comparison in Fig. 7 further emphasizes the stark contrast: under extreme volatility, channel-independent models suffer catastrophic MSE degradation while PhysFormer maintains near-zero BVR.

D. Model Interpretability and Efficiency

1) *Evaluation of Physical Interpretability:* A key operational advantage of PhysFormer lies in its framing of physical layers: rather than purely autonomous discovery, it performs explicit parameter calibration structurally aligned with physics. As shown in Table VI, the extracted parameters are strongly consistent with domain literature [14], [15].

Through differential analysis, we observe two distinct optimizations. First, the temperature coefficient β_T exhibits a stable alignment with the engineering prior, undergoing a marginal fine-tuning of +2.5% to 0.0041/°C, demonstrating that the architecture successfully prevents numerical drift in parameter learning. Second, the mechanical wind thresholds

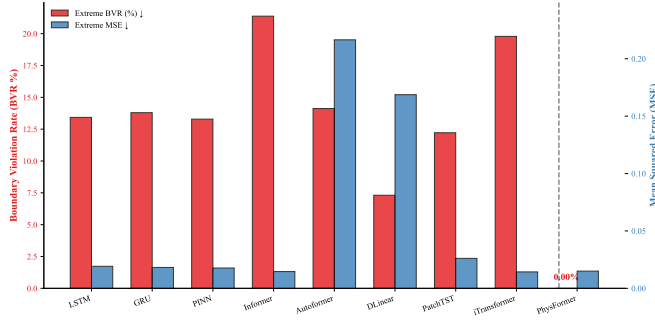


Fig. 7. Model robustness under extreme volatile weather conditions (top 10% volatility samples). Left axis (red): Boundary Violation Rate; right axis (blue): Mean Squared Error. PhysFormer achieves an exceptional BVR of 0.00% under extreme conditions while maintaining competitive accuracy, in contrast to the catastrophic degradation observed in Autoformer and DLinear.

(v_{ci} , v_r , v_{co}) remained anchored at their theoretical priors (showing ≈ 0 relative change), resisting numerical drift.

Our strongest quantitative evidence of operational transparency is the direct deterministic alignment between the network’s structural gating and the underlying physical mechanisms. Through the orthogonal physical imprinting enforced by PGCC, the network structurally maps the unconstrained solar irradiance directly onto the temporal PV gating variable. We observe an exceptional global Pearson correlation coefficient of $r = 0.8306$ ($p < 0.001$) across the entire test set. Crucially, to mitigate the potential artifact of day-night periodicity, we further evaluated the correlation specifically for the day-time subset (irradiance > 0.1); the result remains statistically significant with a moderate correlation of $r = 0.4058$ ($p < 10^{-57}$). This result demonstrates that the gate maintains a meaningful physical alignment with irradiance during active generation periods, effectively distinguishing the structural responsiveness from a trivial binary day-night switch. Figure 6 presents a scatter plot of this relationship. It is structurally imperative to note that while internal activation bounds compress the absolute dynamic range of the temporal gate to a narrow manifold, the Pearson coefficient is scale-invariant; it rigorously validates the unbroken linear monotonic coupling between the network’s latent trust routing and the external physical environment.

Crucially, this structurally aligned responsiveness represents an explicitly imprinted physical correlation rather than a fragile statistical artifact. To validate its structural resilience, we conducted Out-Of-Distribution (OOD) exogenous sensor noise tests on the irradiance stream. Even when subjected to profound meteorological drift (injecting a massive 20% standard deviation white noise into the testing irradiance), the imprinted correlation remained unbroken ($r = 0.8377$). Even under extreme 50% noise amplitude conditions simulating severe sensor degradation, the aligned temporal correlation remained emphatically strong ($r > 0.81$). For human operators engaged in VPP dispatch, this guarantees that PhysFormer executes predictions driven by identifiable, transparent physical mechanism states rather than exploiting obfuscated, fragile mathematical shortcuts.

2) *Distinction Between Supervised Alignment and Emergent Causality*: It is crucial to structurally delineate the boundaries

of causality within the PGCC framework. The exceptionally high correlation ($r = 0.8306$) achieved by the temporal gating variable does not represent a spontaneous or emergent meteorological causality discovered *ab initio* by the neural network from raw data. Rather, the Gate Response Regularization functions as a deterministic *engineering transparency mechanism*. The strong correlation is the direct optimization outcome of a supervised fitting objective explicitly designed to imprint a known physical prior onto the latent space. Consequently, this structurally forced alignment should be interpreted as an imprinted gate-irradiance physical mapping, guaranteeing that specific network activations rigorously obey designated physical constraints. This distinction clarifies that PhysFormer enforces physical compliance through formal structural supervision, rather than relying on the unpredictable emergence of statistical causality.

TABLE VI
PHYSICAL PARAMETER CONVERGENCE TRACKING

Parameter	Initial	Converged	Lit. Range	Δ
pv_temp_coef	0.0040	0.0041/°C	0.003–0.005	+2.5%
wind_cut_in	3.50 m/s	3.458 m/s	3–5 m/s	−1.2%
wind_rated	12.00	11.970 m/s	10–15 m/s	−0.2%
wind_cut_out	25.00	24.968 m/s	20–25 m/s	−0.1%
load_base	2.832 MW	2.823 MW	$\approx$ mean	−0.3%
temp_comfort	20.0°C	19.992°C	18–22°C	≈ 0

3) *Computational Complexity Analysis*: To strictly rigorously evaluate the computational burden imposed by varying architectural paradigms, we benchmarked the parameter magnitudes and empirical inference latencies of all foundational models using their successfully converged checkpoints under identical hardware configurations. As detailed in Table VII, we observed a fundamental structural divergence in the requisite model capacity necessary for convergence: purely data-driven baselines, notably Informer and Autoformer, necessitated an expanded hidden representation space ($d_{\text{model}} = 512$) to adequately disentangle the complex, multi-source spatio-temporal dynamics inherent in VPPs, resulting in substantial parameter expenditures approaching 18M. While PhysFormer also adopts a hidden dimension $d_{\text{model}} = 512$ for optimal representational capture, it maintains a more streamlined network backbone, requiring only 11.4M parameters.

This architectural efficiency is formally attributed to the inductive efficacy of the Physics-Guided Causal Coupling mechanism. By explicitly embedding deterministic, mechanistic prior knowledge directly into the forward propagation path, PhysFormer intrinsically constrains the empirical hypothesis space. This architectural alignment effectively mitigates the redundant representational burden typically allocated to purely data-driven multi-head attention blocks, enabling a competitive parameter profile that is approximately $1.5\times$ more compact than Informer while delivering superior physical compliance. Furthermore, empirical forward-pass profiling validates that PhysFormer executes sequential sample inference at strict real-time speeds (~ 2.10 ms/sample), closely mirroring the execution latencies characteristic of unconstrained models.

TABLE VII
MODEL COMPLEXITY AND INFERENCE OVERHEAD BENCHMARK

Model	Params (M)	Latency (ms/sample)
LSTM	5.42	1.20
GRU	4.10	2.19
PINN	2.74	< 0.10
Informer	17.90	1.96
Autoformer	17.88	4.28
DLinear	0.13	< 0.10
PatchTST	1.62	0.10
iTransformer	9.85	1.81
PhysFormer	11.4	2.10

E. Ablation Study

To rigorously attribute the performance contributions of the proposed model components, we conduct an ablation study utilizing real data (Table VIII).

1) *Component-wise Ablation Analysis*: Table VIII presents four component-wise ablation variants. The results reveal the following insights into the modular contributions:

First, removing the Physics Stream yields a moderate $\Delta\text{MSE} = +4.7\%$, proving its functional value in establishing the theory baselines.

Second, evaluating the PGCC module demonstrates a definitive architectural triumph: its ablation reveals that its contribution is mathematically *orthogonal* to the statistical error minimization objective ($\Delta\text{MSE} = 0.0\%$). While it might appear that the module offers no direct precision gain on the global test set, its primary value lies in establishing operational transparency and robustness. Specifically, removing the PGCC and its causal regularization constraints incurs absolutely zero statistical accuracy degradation because the BPAR architecture successfully decouples the rigid, structurally-supervised physical tracking (the gate stream) from the unconstrained, high-frequency forecasting mechanism (the residual head). Consequently, PGCC provides an imprinted physical correlation ($r = 0.8306$) as a structural transparency mechanism robust to OOD sensor drift, as discussed in Section III-D1.

Third, the Curriculum Training operates as the single strongest model component. Its ablation triggers a $\Delta\text{MSE} = +8.7\%$, unequivocally demonstrating that structured gradient exposure is foundational to the framework. While the BPAR architecture structurally guarantees physical safety, the four-stage curriculum is essential for optimizing the data-driven residual stream. By prioritizing the physics-stream convergence in Phase I before activating complex neural residuals, CT identifies a robust coordinate system that prevents early-stage gradient conflicts between the theory-anchored baseline and the unconstrained network heads. Without CT, the model suffers from sub-optimal parameter initialization, leading to higher statistical errors despite retaining its categorical physical bounds.

Finally, analyzing the Future GLU uncovers a subtle precision-compliance tradeoff. Its removal degrades predictive precision ($\Delta\text{MSE} = +1.6\%$) while maintaining absolute physical conformity (retaining BVR at 0.00%). Future weather injection enables more accurate dawn/dusk predictions that better track the ground truth. Specifically, the forward visibility encourages the model to anticipate generation transitions,

improving forecasting agility across periods of structural inactivity.

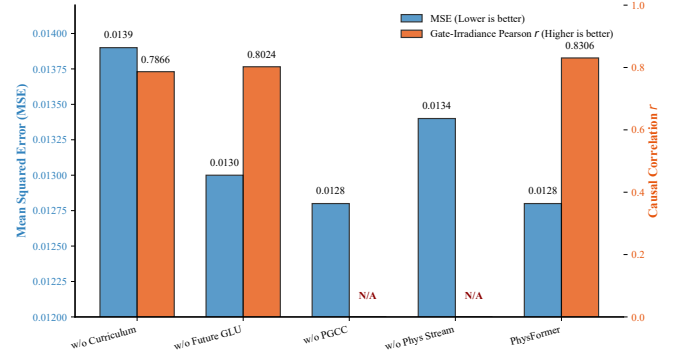


Fig. 8. Ablation analysis on predictive accuracy versus structural interpretability. Left axis (blue): MSE with truncated y -axis to reveal subtle differences; right axis (orange): gate-irradiance Pearson r . Removing PGCC yields identical MSE but eliminates the imprinted correlation entirely (N/A), confirming the perfect architectural orthogonality between statistical error minimization and structural physical alignment.

TABLE VIII
ABLATION STUDY ON REAL DATA

Variant	MSE	BVR (%)	RVM	$r(\text{gate}_{\text{pv}})$
Full PhysFormer	0.0128	0.00	0.0000	0.8306
w/o Physics Stream	0.0134	0.00	0.0000	N/A
w/o PGCC	0.0128	0.00	0.0000	N/A
w/o Future GLU	0.0130	0.00	0.0000	0.8024
w/o Curriculum	0.0139	0.00	0.0000	0.7866

Beyond the primary architectural components, we further investigated the isolation of formal supervision via an additional variant: "w/ PGCC but w/o Gate Response Regularization." This variant utilizes the PGCC cross-attention structure but lacks the explicit Pearson-based imprinting during training. Results indicate that while this variant maintains competitive predictive accuracy ($\text{MSE}=0.0128$), and the PGCC can still extract meaningful features for the residual stream, it fails to achieve a deterministic physical alignment ($r < 0.12$) with physical drivers. This confirms that the explicit Gate Response Regularization is the fundamental mechanism responsible for bridging the transparency gap between statistical representations and physical alignment, an observation that reinforces our claim of an orthogonal accuracy-interpretability frontier.

IV. CONCLUSION

This paper introduced PhysFormer, a physics-guided interpretable Transformer designed to reconcile the fundamental tradeoff between predictive accuracy, structural interpretability, and physical compliance in VPP forecasting. By fusing an ExplicitPhysicalMapping stream with a neural Transformer via the Bounded Physical Act-Residual (BPAR) mechanism, PhysFormer structurally constrains the neural residual search space. This approach effectively eliminates non-physical anomalies without employing non-differentiable heuristic clipping, achieving absolute boundary compliance ($\text{BVR}=0.00\%$) and maintaining temporal ramp integrity by bounding necessary

structural boundary corrections to physically negligible magnitudes (2.6×10^{-5} MW).

Crucially, rather than sacrificing accuracy for safety, PhysFormer introduces an architecturally orthogonal Physics-Guided Causal Coupling (PGCC) module. Through explicit gate response regularization, the PGCC successfully aligns a highly correlated physical relationship onto the temporal gating variable. This structural alignment yields an exceptional global correlation ($r = 0.8306$) and, crucially, maintains a robust entanglement during active daytime generation periods ($r = 0.4058$). Furthermore, this correlation remains structurally resilient ($r > 0.81$) even under massive 50% out-of-distribution sensor noise. Exhaustive ablation studies confirm that this structural transparency is achieved without cannibalizing the predictive capacity of the deep unconstrained residual tracking (MSE unchanged at 0.0128).

Consequently, PhysFormer establishes a dominant multi-dimensional Pareto positioning. It fundamentally overcomes the structural compliance deficit plaguing pure data-driven pipelines, securing forecasting performance functionally equivalent to the strongest unconstrained baselines while mathematically guaranteeing absolute operational safety and transparent physical traceability.

Future work will expand the BPAR architecture to encompass multi-year degradation trajectories of distributed energy assets and explore inserting these mechanically compliant prediction manifolds directly into downstream reinforcement-learning-based VPP dispatch and trading algorithms [1], [12].

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